

# IMAGE PROCESSING TECHNIQUES

*Extended Class Activities Worksheet | UG Level | Activity 9: Edge Detection*

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## WORKSHEET SOLUTIONS

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1. **1. Define Edge Detection in one sentence.**

Edge detection is an image processing technique used to identify boundaries within an image by locating points where pixel brightness or intensity changes sharply.

2. **2. Name any two edge detection operators.**

Sobel operator and Prewitt operator (or Canny / Roberts edge detector).

3. **3. Which operator gives thin and accurate edges?**

The **Canny Edge Detector** gives thin (single-pixel-wide) and accurate edges due to its Non-Maximum Suppression (NMS) stage.

4. **4. Why is Gaussian smoothing used before edge detection?**

Gaussian smoothing reduces high-frequency noise and fine details in an image before gradient computation, preventing noise pixels from being misdetected as false edges (since derivative operations enhance noise).